Arjun Tschand

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Education

Duke University – Durham, NC — Expected May 2028

Bachelor of Science in Electrical and Computer Engineering and Computer Science — **GPA:** 3.83/4.00

Master of Science in Electrical and Computer Engineering (3+1 Program)

- **Coursework**: Deep Learning (Grad), Data Structures & Algorithms, Robust Machine Learning, Natural Language Processing, Probability, Linear Algebra, Multivariable Calculus, Computer Architecture, Signals & Systems

Experience

Incoming Data Science and AI Intern, SoFi — Jun 2026 – Aug 2026

Machine Learning Researcher, Duke University – Zavlanos Lab — Aug 2025 – Present

- Built PyTorch training pipelines applying distributionally robust optimization (GroupDRO) on heterogeneous multimodal datasets, improving worst-group accuracy by 15.0% on the NHANES cardiovascular disease dataset
- Designed a representation learning architecture with per-group encoders and a shared classifier for heterogeneous feature spaces, aligning latent spaces via Gaussian and Wasserstein distance to improve cross-group generalization
- Implemented ETL pipelines, data preprocessing and experiment tooling for hyperparameter & architecture sweeps
- Evaluated on UCI Heart Disease and MNIST/USPS, improving worst-group accuracy up to 8.1% over ERM
- Presented GroupDRO research at Duke ECE poster sessions (Fall 2025, Spring 2026)

Software Engineering Intern, Bloomberg LP — Jun 2025 – Aug 2025

- Developed a Large Language Model (LLM)-powered diagnostic system to automate post-trade issue resolution for fixed income trade allocations, reducing resolution time by 85% on the Sell Side Specialty Assets Team
- Architected a Python and FastAPI backend with LangGraph-based agents that queries 1,200+ trade audit logs per request and leverages Bloomberg's LLMs to identify trade allocation pipeline issues and recommend resolutions
- Designed a React and TypeScript chatbot UI with conversational memory and enterprise SSO authentication
- Implemented agent observability, logging, and evaluation to monitor LLM response quality and system reliability

Software Engineering Intern, Bloomberg LP — Jun 2024 – Aug 2024

- Developed observability tooling for Bloomberg Terminal equity index workflows on the Index Reference Data Team
- Built real-time dashboards visualizing equity index usage metrics of 20,000+ daily users to monitor migration
- Authored flexible Grafana and Humio data queries to power telemetry requests across two internal data systems
- Drove adoption of the new EQIX workflow for 625+ engineers by highlighting legacy-function usage in dashboards

Projects

Crypto Options Pricing & Risk Engine — Apr 2026

- Built BTC/ETH pricer (Black-Scholes, Monte Carlo, Heston), 8.4% MAPE validated on 1,505 Deribit contracts
- Calibrated Heston via differential evolution, outperforming Black-Scholes by 166 pp in 2020/2022 crash regimes
- Designed algorithm to flag 110 contracts with >15% model-vs-market mispricings, generating trading watchlist

Duke Scheduling Assistant — Dec 2025

- Built a Duke course planning and scheduling chatbot using RAG over structured degree and course documents
- Engineered a RAG pipeline: chunked and embedded documents with all-MiniLM-L6-v2, indexed in a ChromaDB vector database, retrieved top-k context via cosine similarity, and generated responses using GPT-4o API
- Evaluated over 100 queries, sweeping chunk size to improve hit@5 from 67% to 86% with 80% end-to-end accuracy

Realtime Drowning Detection — Jan 2024 – Jun 2024

- Built an embedded Raspberry Pi–based system for poolside drowning detection by training a real-time TensorFlow CNN classifier, achieving 96% accuracy on a custom labeled image dataset (drowning, swimming, empty pool)
- Developed a Flask and SQL backend and a Swift iOS app streaming live video with sub-2 second drowning alerts
- Self-filed USPTO provisional patent titled *Intelligent Audio-Visual Observation System to Detect Pool Drownings*

RecyclAIble — Sept 2024

- Collaborated to create computer vision-powered mechanical waste bin, reaching 95.4% classification accuracy
- Won "Best Hardware Hack" out of 62 teams at PennApps XXIII Hackathon

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, Swift, HTML/CSS
Frameworks/Tools: PyTorch, TensorFlow, React, Pandas, NumPy, LangGraph, FastAPI, AWS, Git, Docker, Linux
Interests: DJing, Bollywood movies, Duke Basketball, Fantasy football, Weightlifting, Classical piano